CouncilLens Development Impact Methodology

CouncilLens (CouncilHound) — impact analysis subsystem

July 2026

model version: joint destination–mode Huff, census-block weights

Abstract

CouncilLens produces screening-level impact estimates for proposed local development projects. A deterministic economic module (spatial-interaction retail capture, walking activity, an employment ledger) and a deterministic fiscal module (revenue and service-cost ranges) generate every number; a language model is used only to extract project facts from public documents under a verbatim-quote firewall and to write a narrative that a validator constrains to the structured outputs. This note documents each displayed metric's formula, its academic or statistical basis, the assumptions that drive its uncertainty range, and the implementation decisions behind the current system. The estimates rank likely orders of magnitude under stated assumptions; they are decision-support evidence, not forecasts.

1 Scope and design principles

1. **Deterministic quantities.** All numbers are produced by closed-form models over open data. The language model never supplies a quantity: extracted project facts require a verbatim source quote ($\leq$15 words) that is verified against the document text in code, and fields without one are nulled. The narrative report is post-validated so that every number it contains matches a published metric, a stated assumption, a provenance string, or a deterministic module note.
2. **Assumption transparency.** Every contestable parameter is a named assumption with a central value, low/high bounds, and a recorded basis. Bounds propagate through all formulas by interval arithmetic, so each metric carries a range derived from the same assumptions it cites.
3. **Fail loudly.** Missing inputs (an unpinned tax rate, an absent assessment record) degrade the affected metric to "not computed — reason," never to a silent default.
4. **Screening honesty.** Capture, walking, and foot-traffic figures are screening estimates in the spatial-interaction tradition; the report states this in plain language and names the most sensitivity-driving assumptions.

2 Notation and data inputs

U	proposed dwelling units (extracted from project documents)
ρ	occupancy rate at stabilization (`occupancy_rate`)
$\bar{s}$	persons per occupied multifamily unit (`avg_hh_size_multifamily`)
y	mean household income of the site's census tract (ACS B19025/B11001)
π	new-construction income premium (`income_premium_new_construction`)
e_c	CES average annual expenditure for category c (BLS CE 2023)
y_{CES}	CES average pre-tax income per consumer unit ($101,805, 2023)
ε_c	expenditure–income elasticity for category c (Engel scaling)
κ	CES scaling factor (`ces_scale`)
A_j	attractiveness of destination j (POI count = 1 per business)
$t_j^w,\ t_j^d$	walk / drive travel time (minutes) from the site to j
$\beta_w,\ \beta_d$	walk / drive impedance decay per minute
w_c	zero-impedance walk preference for category c
τ	daily walk trips per resident (`walk_trips_per_resident_day`)

Networks are OpenStreetMap walk and drive graphs built with OSMnx [9]; walk edge times use 4.8 km/h. Population weights use 2020 Census (PL 94-171) block centroids. Points of interest merge Overture Maps, OpenStreetMap, and (when licensed) Foursquare OS Places into a nine-class taxonomy. Travel times come from single-source Dijkstra runs over edge minutes; each business inherits the time of its nearest network node.

3 Demand module

3.1 New households and residents

$$N_{hh} = U \cdot \rho \tag{1}$$

$$N_{res} = N_{hh} \cdot \bar{s} \tag{2}$$

Occupancy at stabilization is benchmarked against the Census Housing Vacancies and Homeownership series [15]. Household size uses a multifamily bedroom-mix demographic multiplier in the tradition of Listokin et al. [12] ($\bar{s} = 1.8$, range 1.5–2.2, for a typical studio/1BR/2BR mix): the ACS B25010 tenure-level renter average (≈ 3.1 locally) was evaluated and rejected as an estimator because it is dominated by single-family renter households and is blind to unit mix. When a project's bedroom mix is known, the multiplier should be set from it (§10).

3.2 Aggregate income and category spending

$$Y = N_{hh} \cdot y \cdot \pi, \qquad r = \frac{y\,\pi}{y_{\text{CES}}} \tag{3}$$

$$S_c = N_{hh} \cdot e_c \cdot r^{\,\varepsilon_c} \cdot \kappa \tag{4}$$

The tract income y is the *mean* (ACS aggregate household income B19025 divided by households B11001, summed over the site tract's block groups) — the correct estimator for total purchasing power, since median $\times$ count understates aggregate income under right-skewed distributions.

When the aggregate tables are suppressed the model falls back to the population-weighted median and records the substitution. Category spending uses BLS Consumer Expenditure Survey 2023 levels [13] scaled by the income ratio r raised to a per-category expenditure elasticity ε_c (grocery 0.45, restaurant/bar 0.85, comparison 0.95, convenience 0.60, personal services 0.90, entertainment 1.05), consistent with the Engel gradients in the CE income-quintile tables: necessity spending grows sublinearly with income, so a high-income tract no longer implies proportionally more grocery spending.

4 Retail capture: joint destination–mode choice

Retail capture follows the probabilistic trade-area tradition of Huff [1, 2] with the exponential impedance of the entropy-maximizing spatial-interaction family [3]. Destination and mode are chosen *jointly*, as a multinomial logit [4, 5] over (business, mode) pairs. For category c, business j, and mode $m \in \{w, d\}$:

$$P_c(j,m) \;=\; \frac{A_j^{\alpha}\,\omega_{c,m}\,e^{-\beta_m t_j^m}}{\sum_k A_k^{\alpha}\left(\omega_{c,w}\,e^{-\beta_w t_k^w} + \omega_{c,d}\,e^{-\beta_d t_k^d}\right)}, \qquad \omega_{c,w} = w_c, \quad \omega_{c,d} = 1 - w_c, \tag{5}$$

with $\alpha = 1$. The mode preference w_c has a clean interpretation: at zero travel time for both modes, the walk share is exactly w_c; as walk time grows, walking loses utility to driving and the trip (and its dollars) arrives by car instead. Annual capture and walk-arriving capture at business j:

$$C_j = \sum_c S_c \sum_m P_c(j,m), \qquad C_j^{walk} = \sum_c S_c\, P_c(j,w). \tag{6}$$

Two invariants are pinned by unit tests: $\sum_j C_j = \sum_c S_c$ (joint probabilities sum to one), and $\sum_j C_j^{walk} \le \sum_c S_c w_c$ with equality only when every destination is at zero travel time — the walk total is *endogenous*, not an assumption restated.

Own-retail destination. The project's ground floor competes inside the model rather than being appended outside it. Its retail square footage is converted to POI-count-equivalent attractiveness (A_{own} = sqft/`own_retail_sqft_per_equiv_poi`), distributed across a fixed category mix (50% restaurant/bar, 15% comparison, 15% convenience, 20% personal services), at zero travel time from the site. Huff's original attractiveness variable was selling area [1]; competitor attractiveness here is POI count because open POI data lacks floor area, so the own-retail conversion reconciles the two units.

In-city shares. For food-away and for all-retail, the in-city capture share is the probability mass at destinations inside the city boundary, $\theta = \sum_{j \in \text{city}} \sum_m P(j,m)$. These shares are the cross-module inputs to the meals-tax and sales-tax lines.

Reporting rollup. Businesses are grouped for display by DBSCAN [8] ($\varepsilon = 150\,\text{m}$, minPts $=$ 3) and each cluster is labeled by the named street intersection nearest its centroid (from the OSM walk graph, within $\sim$450 m), falling back to the central member business where no named intersection is close; cluster values are sums of member-business captures and play no role in the allocation itself.

5 Walking activity

5.1 Resident walk trips and exact marginal flows

Daily walk trips are $T = N_{res} \cdot \tau$. These all-purpose trips are allocated over POI-weighted walk-network nodes with exponential decay and routed along shortest paths — deterministically, with no sampling — giving per-street-segment flows:

$$f_e \;=\; \sum_v T \cdot \frac{g_v \, e^{-\beta_w t_v}}{\sum_u g_u \, e^{-\beta_w t_u}} \cdot \mathbb{1}[e \in \mathrm{SP}(\mathrm{site} \to v)]\,, \tag{7}$$

where g_v is the POI weight at node v and SP is the shortest path. The change caused by the project *is* the trips its residents generate, so no baseline differencing (and no sampling noise) enters the numerator.

5.2 Baseline index and percent comparison

The comparison baseline is a seeded, sampled, weighted betweenness index [6, 7] of today's walkers: $k = 2000$ origin–destination pairs drawn with origins proportional to census-block population within 400 m of each node and destinations proportional to POI weight, routed by shortest paths. Scaling the sampled counts by (total population weight $\times\, \tau)/k$ expresses the baseline in the same trips units, and the trip rate τ cancels from the percentage:

$$\Delta_e \;=\; 100 \cdot \frac{f_e}{n_e \cdot \bar{P}\,\tau/k} \quad \text{(reported only where } n_e \geq 5\text{)}, \tag{8}$$

with n_e the sampled baseline count on edge e and $\bar{P}$ the total population weight. The headline metric averages Δ_e over the ten commercial street segments nearest the site. Without calibrated pedestrian counts this is an index comparison, not a volume estimate; the correlation of betweenness-type indices with observed pedestrian volumes is supported by the network-analysis literature but is context-dependent.

5.3 Consistency diagnostic

$$\text{spending per walk trip} = \frac{\sum_j C_j^{walk}}{365\,T}. \tag{9}$$

Because T counts *all* walking (exercise, transit access, dog walks) while the numerator is shopping only, a plausible value is a few dollars per average trip (implying \$5–\$15 per actual shopping trip). Values outside \$0.25–\$30 add a consistency flag to the report.

6 Employment ledger

$$J^- = \frac{\text{existing commercial sqft}}{\texttt{sqft\_per\_office\_job}}, \qquad J^+ = \frac{\text{proposed retail sqft}}{\texttt{sqft\_per\_retail\_job}}, \qquad \Delta J = J^+ - J^-. \tag{10}$$

Space-per-worker values (300 [200–450] office; 500 [400–700] retail, gross) sit between dense workplace-utilization benchmarks and the building-stock averages observable in CBECS [16]. These are site-activity screens only; no indirect or induced employment is modeled.

7 Fiscal module

The module reports both of the two canonical fiscal-impact framings — average(per-capita) cost and marginal cost — following Burchell & Listokin's classification [10] and the planning literature's standard caution that neither is universally correct for infill [11].

7.1 Revenue

$$\text{current RE tax} = AV_{now} \cdot r/100 \tag{11}$$

$$AV_{proj} = \text{median}\left(\frac{AV_i}{\text{units}_i}\right) \cdot U \;+\; \text{retail sqft} \cdot p_{sqft} \qquad \text{(bounds: comp 25th–75th pct)} \tag{12}$$

$$\text{projected RE tax} = AV_{proj} \cdot r/100 \tag{13}$$

$$\text{RE tax increase} = (AV_{proj} - AV_{now}) \cdot r/100 \tag{14}$$

$$\text{meals tax} = S_{\text{food away}} \cdot \theta_{\text{food}} \cdot r_{\text{meals}} \tag{15}$$

$$\text{local sales tax} = (S_{\text{grocery}} + S_{\text{comparison}} + S_{\text{convenience}}) \cdot \theta_{\text{all}} \cdot r_{\text{sales}} \tag{16}$$

$$\text{personal property} \approx N_{hh} \cdot \nu_{hh} \cdot \bar{v}_{veh} \cdot r_{pp}/100 \qquad \text{(rough estimate)} \tag{17}$$

$$\text{BPOL} \approx \text{retail sqft} \cdot g_{sf} \cdot r_{bpol}/100 \qquad \text{(rough estimate)} \tag{18}$$

Comparables are apartment-class parcels (land-use code 352) built within a 15-year lookback, pulled from the city's public assessment database with per-unit values AV_i/units_i; fewer than three comps degrades the metric with a low-confidence flag. All rates (r, r_{meals}, r_{sales}) are jurisdiction configuration pinned with source URL and fiscal year. Virginia's 1% local-option sales tax applies to food for home consumption, so grocery is correctly in the taxable base; personal services are excluded as generally non-taxable.

Personal property and BPOL revenue are *rough estimates*, and are labeled as such wherever they appear. Their tax rates are pinned city policy — r_{pp} = \$4.13 per \$100 of vehicle value and r_{bpol} = \$0.20 per \$100 of retail gross receipts, both from the city's adopted-budget Rates & Levies schedule — but the bases are assumptions: vehicles per household (ν_{hh}), average vehicle value ($\bar{v}_{veh}$), and tenant gross sales per square foot (g_{sf}) have no project-specific data behind them and carry wide bounds. When per-household personal-property budget actuals (p_{pp}) are pinned, the actuals path $N_{hh} \cdot p_{pp}$ replaces Equation (17). Both estimate lines enter incremental recurring revenue: omitting them would systematically understate revenue for taxes the city does levy. (PPTRA car-tax relief is a fixed state block grant, so marginal vehicles yield the city the full levy; the services BPOL classification at \$0.27 per \$100 is slightly above the retail rate applied here.)

7.2 Service costs and net impact

When the city's education transfer E and school enrollment N_{enr} are pinned, costs use the *school-split* model — the per-capita-multiplier refinement of [10]: school costs follow the project's own student estimate at the per-pupil tuition rate, and only non-school costs are allocated per resident.

$$S_{K12} = U \cdot \texttt{students\_per\_unit} \tag{19}$$

$$\text{naive cost} = N_{res} \cdot \frac{GF - E}{P} \; + \; S_{K12} \cdot \frac{E}{N_{enr}} \tag{20}$$

$$\text{marginal cost} = N_{res} \cdot \frac{GF - E}{P} \cdot \mu \; + \; S_{K12} \cdot \frac{E}{N_{enr}}, \qquad \mu = \texttt{marginal\_cost\_factor} \tag{21}$$

$$\text{net}_{\,method} = R_{inc} - \text{cost}_{\,method} \tag{22}$$

where R_{inc} is *incremental* recurring revenue (the real-estate component is the tax *increase* when the current tax is known). The school term appears unscaled in both framings: pupils generate tuition-contract costs one-for-one, so a development generating few students carries proportionally low school costs instead of the school-heavy citywide average — without the split, a zero-student building would still be billed average school costs for every resident. The student generation rate is set toward the high-rise/small-unit end of the multifamily evidence [12] because university-adjacent rental buildings house predominantly single and roommate households rather than families. The naive framing remains an upper bound on the non-school side (it allocates fixed citywide costs to incremental residents); the marginal framing scales only the non-school share expected to grow and can understate costs if capacity thresholds are crossed. The headline range spans both framings; per-method nets are also published. If E or N_{enr} is unpinned, both framings degrade to the single per-capita form $N_{res} \cdot GF/P$ with the student count reported as a note.

8 Assumptions

Central values with propagated bounds; every metric lists the assumption keys that touched it.

Key	Value	Range	Basis
`occupancy_rate`	0.95	0.90–0.97	stabilized MF occupancy; cf. HVS vacancy [15]
`avg_hh_size_multifamily`	1.8	1.5–2.2	bedroom-mix multipliers [12]
`income_premium_new_construction`	1.125	1.0–1.25	screening premium, stated openly
`ces_scale`	1.0	0.85–1.15	CES national-average drift allowance [13]
`beta_walk` (min^{-1})	0.15	0.07–0.15	impedance decay; set to top of band (2026-07)
β_d (min^{-1}, fixed)	0.15	—	drive impedance decay
`walk_share_neighborhood`	0.60	0.40–0.80	zero-impedance walk preference
`walk_share_comparison`	0.10	0.00–0.30	comparison goods travel by car
`walk_share_grocery_entertainment`	0.30	0.10–0.50	interpolated
`walk_trips_per_resident_day`	1.0	0.5–2.0	NHTS metro rates + walkability uplift [14, 17]
`sqft_per_office_job`	300	200–450	planning standard; cf. CBECS [16]
`sqft_per_retail_job`	500	400–700	planning standard; cf. CBECS [16]
`own_retail_sqft_per_equiv_poi`	2,000	1,500–3,000	typical inline suite size
`commercial_value_per_sqft`	\$275	\$200–\$400	screening; replace with comps
`students_per_unit`	0.10	0.05–0.15	high-rise MF, univ.-adjacent [12]
`marginal_cost_factor`	0.40	0.25–0.60	share of average cost that scales
`vehicles_per_household`	1.4	1.0–1.8	ACS vehicle norms, MF renters (estimate)
`avg_vehicle_assessed_value`	\$14,000	\$9k–\$20k	NoVA personal-property rolls (estimate)
`retail_sales_per_sqft`	\$400	\$250–\$600	neighborhood retail gross sales (estimate)

Structural constants: walk speed 4.8 km/h; DBSCAN $\varepsilon = 150\,\text{m}$, minPts = 3; network extent = city boundary + 3 km; POI study area = boundary + 2 mi; node population radius 400 m; baseline sampling $k = 2000$, fixed seed; comp lookback 15 years.

9 Implementation decisions

- **Per-business destinations.** The Huff run treats every retail POI as an individual destination (a single-source Dijkstra makes per-POI times free); DBSCAN clusters exist only as reporting rollups. Capture is therefore spatially resolved to business locations, and map heat layers consume real per-business weights rather than interpolated cluster centroids.
- **Joint rather than nested choice.** Equation (5) is a single multinomial logit over (destination, mode) pairs. A nested logit would relax the independence-of-irrelevant-alternatives property across modes; the joint form was chosen for transparency and because screening bounds on w_c and β_w dominate the error budget.
- **Exact marginal flows.** Project-caused foot traffic is computed as the shortest-path routing of the residents' own trips, replacing an earlier sampled before/after betweenness difference whose noise exceeded the signal at this project scale. Sampling now appears only in the baseline denominator.
- **Census-block population.** Node population weights use 2020 census block centroids (the finest published geography) fetched by spatial envelope, which also extends the baseline beyond the city line that the county-scoped ACS layer stopped at.
- **Determinism.** All sampling is seeded; two evaluations on a warm cache produce identical metric values.
- **LLM firewall.** Document extraction demands a verbatim $\leq$15-word quote per numeric field, verified in code against the document text; unverifiable values are nulled. Parcel identifiers are verified the same way.
- **Narrative validator.** The generated report is scanned for numbers; each must match a metric value/bound, spec quantity, assumption, or deterministic note within rounding, else the draft is regenerated once and then hard-fails.
- **Assessment data.** The city publishes no bulk assessment layer, so site values and multifamily comparables come from the public Patriot WebPro search interface, throttled to $\sim$1 request/s and cached; owner names are neither parsed nor stored.
- **Map scales.** Heat layers are clipped to commercial-retail zoning polygons and colored on dollar-pinned viridis ramps (nice-number ceilings), so a color means the same dollars regardless of layer toggling; capture and walk-in capture share one scale to make their comparison meaningful.
- **Interactive assumption adjustment.** Each metric publishes an exact power-law decomposition over the assumptions it depends on multiplicatively (or by a fixed power, e.g. the Engel elasticities): $v' = \sum_t v_t \prod_k (a_k/a_k^0)^{e_{t,k}}$. Because the joint-choice probabilities are independent of the demand-side assumptions, this client-side recompute reproduces the pipeline's own arithmetic exactly; the evaluation asserts $\sum_t v_t = v$ on every run. Network-model parameters (β_w, mode shares, own-retail attractiveness) appear in no term and are presented as requiring a full re-run.

10 Known limitations and audit notes

Four limitations identified in the July 2026 audit have been addressed in this version; their adopted resolutions and residual caveats:

1. **Household size (addressed).** The unit-mix-blind ACS renter average (≈ 3.1) was replaced by a multifamily bedroom-mix multiplier, $\bar{s} = 1.8$ [1.5–2.2] [12]. *Residual:* the multiplier assumes a typical studio/1BR/2BR mix; when a project's actual bedroom program is extracted it should set $\bar{s}$ directly.

2. **Aggregate income (addressed).** Tract income now uses the mean (B19025/B11001) rather than median $\times$ households, with a median fallback under suppression. *Residual:* tract means still smooth over building-level differences; the new-construction premium carries that uncertainty.

3. **CES income scaling (addressed).** Category spending scales by r^{ε_c} per Equation (4) instead of linearly. *Residual:* the elasticities are national approximations read from CE quintile gradients, not locally estimated.

4. **Walk-trip rate (addressed).** τ reduced from 2.0 to 1.0 [0.5–2.0], anchored to NHTS metro person-level walking rates with a walkability uplift consistent with the built-environment literature [14, 17]. The foot-traffic *percentage* is unaffected (τ cancels) and walk-in dollars are independent of τ; trips and street-flow magnitudes scale with it. *Residual:* local counts remain the real calibration.

5. **IIA.** The joint logit imposes proportional substitution across all (destination, mode) pairs; a nested structure would be more standard.

6. **One-sided β_w band.** The central value sits at the top of its sensitivity range (analyst calibration), so sensitivity sweeps only explore gentler decay.

7. **Attractiveness proxy.** Competitor attractiveness is POI count (floor area being unavailable in open POI data), while own-retail attractiveness is derived from square footage; the conversion constant carries the reconciliation and its bounds.

8. **Sales-tax base share.** The taxable base applies the all-retail in-city share to the three taxable categories; a category-specific share would be marginally more precise.

9. **Foot-traffic index is uncalibrated.** Percent changes compare modeled flows to a sampled opportunity index, not to counted pedestrians; local counts would convert the layer to volumes.

10. **Fiscal method dependence.** The net range spans average-cost and marginal-cost framings on purpose; the midpoint of that range is a presentation convenience, not an estimate with independent standing.

11. **Rate-based revenue estimates.** Personal property and BPOL revenue use pinned city rates but assumed bases, per Equations (17) and (18); they are labeled rough estimates and carry the widest relative bounds on the revenue side. Per-household personal-property budget actuals and tenant-level receipts would replace them.

11 Data sources

City of Fairfax GeoHub ArcGIS services (boundary, parcels, zoning); City of Fairfax public assessment database (Patriot WebPro); City budget pages (tax rates, General Fund, pinned with fiscal year) and the adopted-budget Rates & Levies schedule (personal property and BPOL rates); City of Fairfax Schools FY2026 budget announcement (FCPS tuition contract and projected ADM, for the school-split cost model); Census ACS 5-year (B01003, B19013, B25010, B25044, B08301) and 2020 PL 94-171 block population via TIGERweb; LEHD LODES 8; OpenStreetMap via OSMnx [9]; Overture Maps places; BLS Consumer Expenditure Survey 2023 [13]; FHWA National Household Travel Survey [14]; Virginia Code §58.1-605 (local-option sales tax).

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